

# SatQuery-Edge

Offline Satellite Intelligence Investigation Report  
SIH 26167 · ISRO · Prototype · Offline Mode

|                  |                               |
|------------------|-------------------------------|
| Investigation ID | SQE-TEST-001                  |
| Timestamp        | 2026-09-02 19:40:29           |
| Intent           | bi_temporal_change            |
| Tools Used       | change, grounding, geospatial |

## Query

Test query

## Final Answer

Test answer

## Evidence Score

| Component            | Code | Value      | Weight |
|----------------------|------|------------|--------|
| Model Agreement      | M    | 0.876      | 0.30   |
| Sensor Reliability   | S    | 0.900      | 0.20   |
| Spatial Consistency  | P    | 0.947      | 0.20   |
| Temporal Consistency | T    | 0.940      | 0.15   |
| Input Quality        | I    | 0.617      | 0.15   |
|                      |      |            |        |
| FINAL SCORE          |      | 86.6 / 100 |        |

## Validation

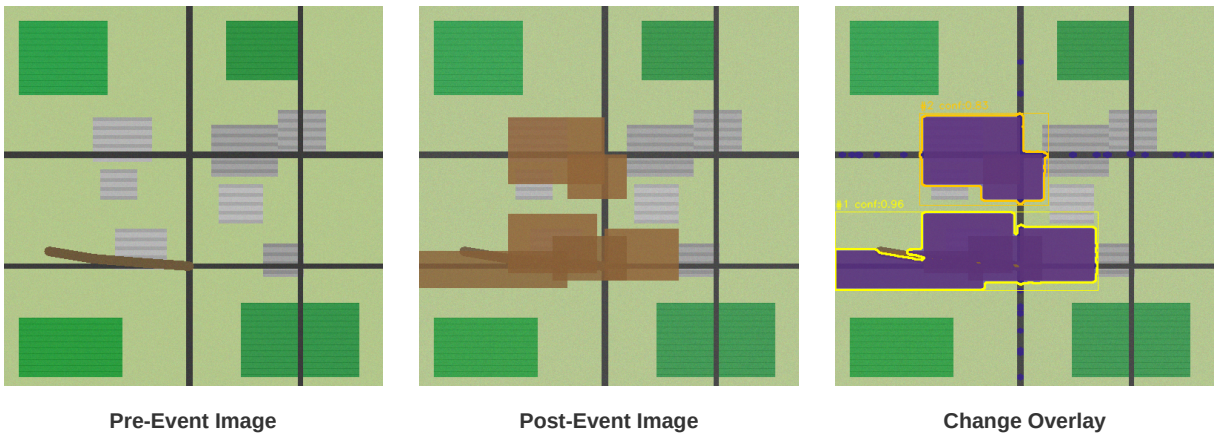
Status: **PASS** ✓

- 2 image(s) detected — count OK.
- Image 1: Quality OK (512×512, 3 channel(s), mean=163, std=39.3).
- Image 2: Quality OK (512×512, 3 channel(s), mean=158, std=37.1).
- Image pair: identical dimensions — perfect pixel grid alignment.

## Detected Regions & Approximate Coordinates

| # | Label                    | Confidence | Area (km²) | Demo Georeferenced |
|---|--------------------------|------------|------------|--------------------|
| 1 | candidate changed region | 0.96       | 18.215     | Yes                |
| 2 | candidate changed region | 0.83       | 9.886      | Yes                |

## Key Visual Evidence



## Specialist Tool Outputs

### CHANGE — SUCCESS — 2 region(s)

Detected 2 significant candidate changed region(s).

Largest region:  
Approximate change area: 10.3% of scene.  
Spectral Change Confidence: 0.96  
Center: (0.37, 0.66) (normalized pixel space)

Total changed pixels: ~17.3% of observation area.

Remote Sensing Analysis (NDWI / Inundation Dynamics):  
Candidate newly inundated / flooded regions where spectral reflectivity changed significantly b

### GROUNDING — SUCCESS — 0 region(s)

No specific target feature found matching 'water'. Please specify target keywords (e.g. water, flood, building, road, vegetation).

### GEOSPATIAL — SUCCESS — 0 region(s)

Geospatial conversion complete.  
Converted 2 region(s) to geographic coordinates.  
Total approximate area: 28.10 km².

■ Location Georeferenced to: Bhubaneswar, Odisha (Default Demo)  
Center Coords: (20.2960°N, 85.8240°E)  
Approximate pixel-to-geographic affine scale applied.

## Execution Trace

## Tool Versions

| Component               | Version        |
|-------------------------|----------------|
| SatQuery-Edge Prototype | 1.0.0-SIH26167 |
| Python OpenCV           | 4.11.0         |
| NumPy                   | 1.26.4         |
| Streamlit               | 1.62.0         |

|               |                            |
|---------------|----------------------------|
| Report Engine | ReportLab PDF              |
| Mode          | OFFLINE — No external APIs |

NOTE: This prototype uses lightweight local computer vision and synthetic/demo geospatial observations where applicable. Results are intended for demonstration purposes only and are NOT scientifically validated remote-sensing products. No cloud APIs, external inference services, or internet connectivity is used. All processing is fully offline. —  
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